AmirPouya MOEINI

PhD Student, University of Cambridge, Trinity Hall

sam297@cam.ac.uk pouya-moeini pouyamoeini.github.io

Education

PhD in Engineering (Information Engineering) Oct 2023 – Present

University of Cambridge, UK | Probabilistic Systems, Information, and Inference Group (PSI^2)
Supervisor: Prof Albert Guillén i Fàbregas
Adviser: Prof Ramji Venkataramanan

BSc in Computer Engineering and Mathematics Sep 2018 – Jun 2023

Sharif University of Technology, Iran
Thesis: *Information-Theoretic Bounds for Online Learning and Sequential Prediction*
Supervisor: Prof Mahdieh Soleymani Baghshah

Research Profile

PhD research: I study the mathematical foundations of Shannon theory, establishing what reliable communication can and cannot achieve when the decoder does not know the channel, either because it uses a mismatched metric or because it must work for every channel in a class (universal decoding). My main tools are large deviations, high-dimensional probability and probabilistic combinatorics. Outcomes include a proof that the expurgated exponent is not universally achievable, answering a question posed by Csiszár and Körner in 1981 [J1].

Broader interests: Developing probabilistic and information-theoretic techniques for inference and random structures problems, and using them to find the limits of what can be learned from data, especially when the model is misspecified.

Publications

Under review

[S3] Universal Decoding for Unequal Error Protection arXiv
A. Moeini and A. Guillén i Fàbregas, submitted to *IEEE Information Theory Workshop (ITW)*, 2027.

[S2] An Upper Bound to the Undetected Error Exponent with Erasure Decoding at Positive Rates arXiv
A. Moeini, M. Dalai, and A. Guillén i Fàbregas, submitted to *IEEE Information Theory Workshop (ITW)*, 2027.

[S1] A Counterexample to the Universal Achievability of the Expurgated Error Exponent
A. Moeini, M. Dalai, and A. Guillén i Fàbregas, submitted to *IEEE Information Theory Workshop (ITW)*, 2027.

Journal papers

[J1] The Expurgated Error Exponent is Not Universally Achievable arXiv
A. Moeini, M. Dalai, and A. Guillén i Fàbregas, to appear in *IEEE Transactions on Information Theory*.

Conference papers

[C4] Random Gilbert–Varshamov Codes for Joint Source–Channel Coding IEEE Xplore
A. Moeini and A. Guillén i Fàbregas, *IEEE International Symposium on Information Theory (ISIT)*, 2026.

[C3] Two-Class Joint Source–Channel Coding: Expurgated Exponents with i.i.d. Distributions DOI
A. Moeini and A. Guillén i Fàbregas, *International Zurich Seminar on Information and Communication (IZS)*, 2026.

[C2] Dual-Domain Exponent of Maximum Mutual Information Decoding IEEE Xplore
A. Moeini and A. Guillén i Fàbregas, *IEEE Information Theory Workshop (ITW)*, 2025.

[C1] Class-Based Expurgation Attains Csiszár's Expurgated Source–Channel Exponent IEEE Xplore
A. Moeini and A. Guillén i Fàbregas, *IEEE Information Theory Workshop (ITW)*, 2025.

Honours and Awards

2023 **Cambridge International Full Scholarship** Cambridge, UK
2022 **Summer Research Fellowship** *Aalto University* Espoo, Finland
2018 **Gold Medal** *International Olympiad on Astronomy and Astrophysics* Beijing, China
2017 **Gold Medal** *National Olympiad on Astronomy and Astrophysics* Iran

Academic Visits

Universitat Politècnica de Catalunya (UPC), Barcelona, Spain Oct 2026
Department of Signal Theory and Communications

University of Brescia, Brescia, Italy Sep 2025, Sep 2026
Department of Information Engineering, hosted by Prof Marco Dalai

National University of Singapore, Singapore Oct 2025
Institute of Data Science – School of Computing, hosted by Prof Jonathan Scarlett

Teaching Experience

Supervisor 🏛 University of Cambridge Oct 2024 – Present

- Information Theory and Coding (3F7, 3rd year), Michaelmas 2024–26
- Data Transmission (3F4, 3rd year), Lent 2025, 2026
- Mathematical Methods (2P7, 2nd year), 2026–27
- Mathematics (1P4, 1st year), Easter 2026, Michaelmas 2026

Teaching Assistant 🏛 Sharif University of Technology 2020 – 2023

Information Theory and Coding, Theoretical Machine Learning, Statistical Machine Learning, Artificial Intelligence, Mathematical Statistics, Regression Analysis, Linear Algebra, Data Structures and Algorithms.

Work Experience

Research Fellow | *Probabilistic Machine Learning Group, Aalto University* Jul 2022 – Oct 2022

Supervised by Prof Samuel Kaski. Worked on making Approximate Bayesian Computation (ABC) robust to outliers and to uninformative or misspecified summary statistics. Developed a gradient-free method that weights the summary statistics by minimising the maximum mean discrepancy (MMD), favouring robust statistics such as the median over outlier-sensitive ones such as the maximum, and a sampler that keeps the robustness of MMD while using only a small budget of MMD evaluations.

Data Scientist Intern | *Comma AI-Med* Mar 2023 – Jun 2023

Developed a pipeline based on generative adversarial networks (GANs) that synthesises realistic tumours in mammography images. The synthetic images were used to augment the scarce labelled data available for training tumour-detection models.

Data Engineer Intern | *Institute for Research in Fundamental Sciences (IPM)* Jun 2020 – Sep 2020

Built a data pipeline for weather-station measurements from the Iranian National Observatory (INO), and trained a model on these measurements to predict cloud cover at the site, a key factor in scheduling telescope observations.

Selected Courses

University of Cambridge: Computational statistics and machine learning, Advanced information theory and coding.

Sharif University of Technology: Information theory and coding, Information-theoretic methods in high-dimensional statistics, High-dimensional probability, Graphical models, Bayesian methods in statistics, Machine learning theory.

University of Cambridge (audited): Probabilistic combinatorics, Entropy methods in combinatorics, Ramsey theory, Random discrete structures, The Ising model, Percolation, Stochastic calculus and applications to finance.

Skills

Programming: Python, C++, SQL, Julia, R, Java

Scientific computing and ML: PyTorch, NumPy, pandas, SciPy, scikit-learn

Tools: Git, Linux, LaTeX

Service and Memberships

Reviewer: *IEEE Transactions on Information Theory*; IEEE International Symposium on Information Theory (ISIT); International Zurich Seminar on Information and Communication (IZS).

Member: IEEE Information Theory Society; Cambridge Philosophical Society; The Archimedeans.

References

Prof Albert Guillén i Fàbregas
Professor, University of Cambridge
✉ albert.guillen@eng.cam.ac.uk

Prof Marco Dalai
Associate Professor, University of Brescia
✉ marco.dalai@unibs.it